AI-Based E-Waste Management Tracker and Analytics Dashboard

A Case Study on the State of Himachal Pradesh



Software Engineering Project

Team: Mayur Jadhav, Soham Gaikwad, Prajwal Bhosale

TE CMPN B

Objective: To monitor and evaluate e-waste management performance using operational data, KPIs, automated scoring, and an analytics dashboard.

Gemini Notebook

Replacing fragmented reporting with automated data pipelines

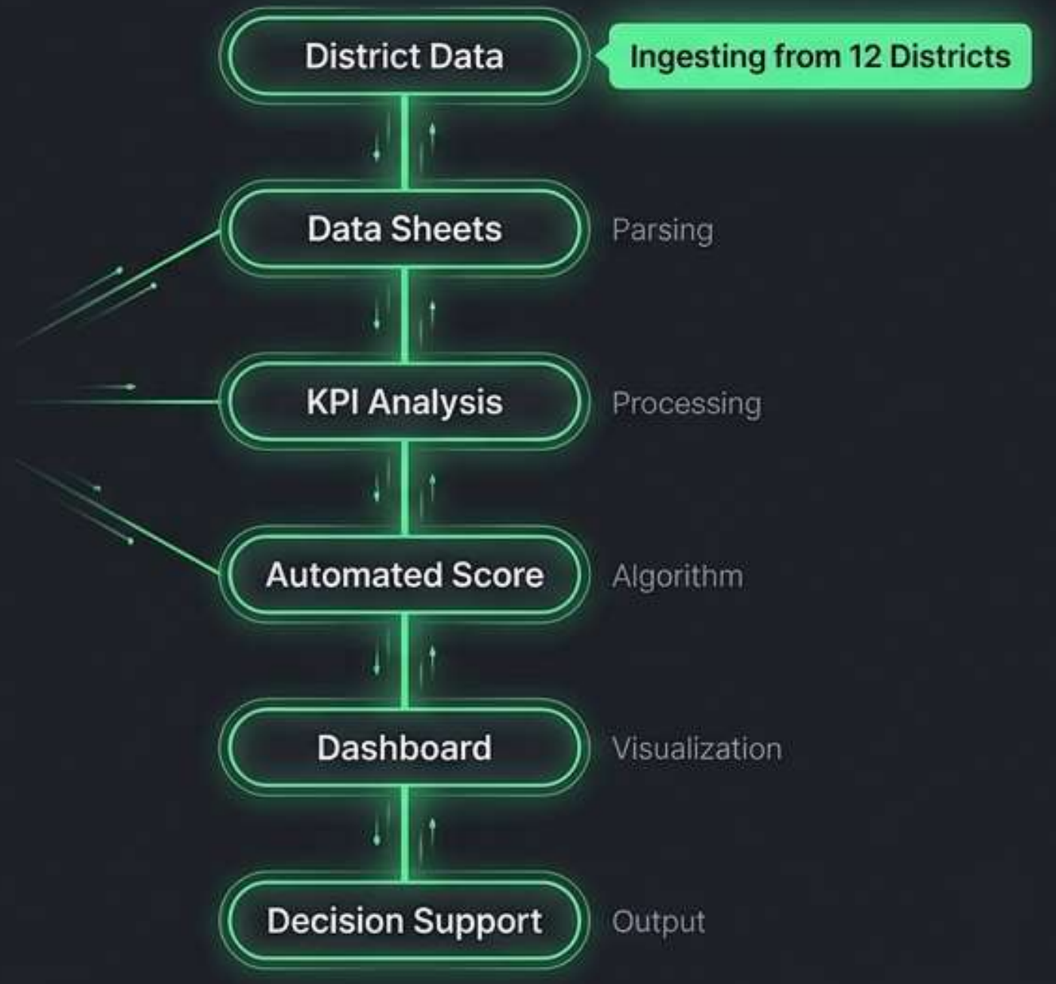
The Problem

Increasing Generation

Siloed Stakeholders & Fragmented Records

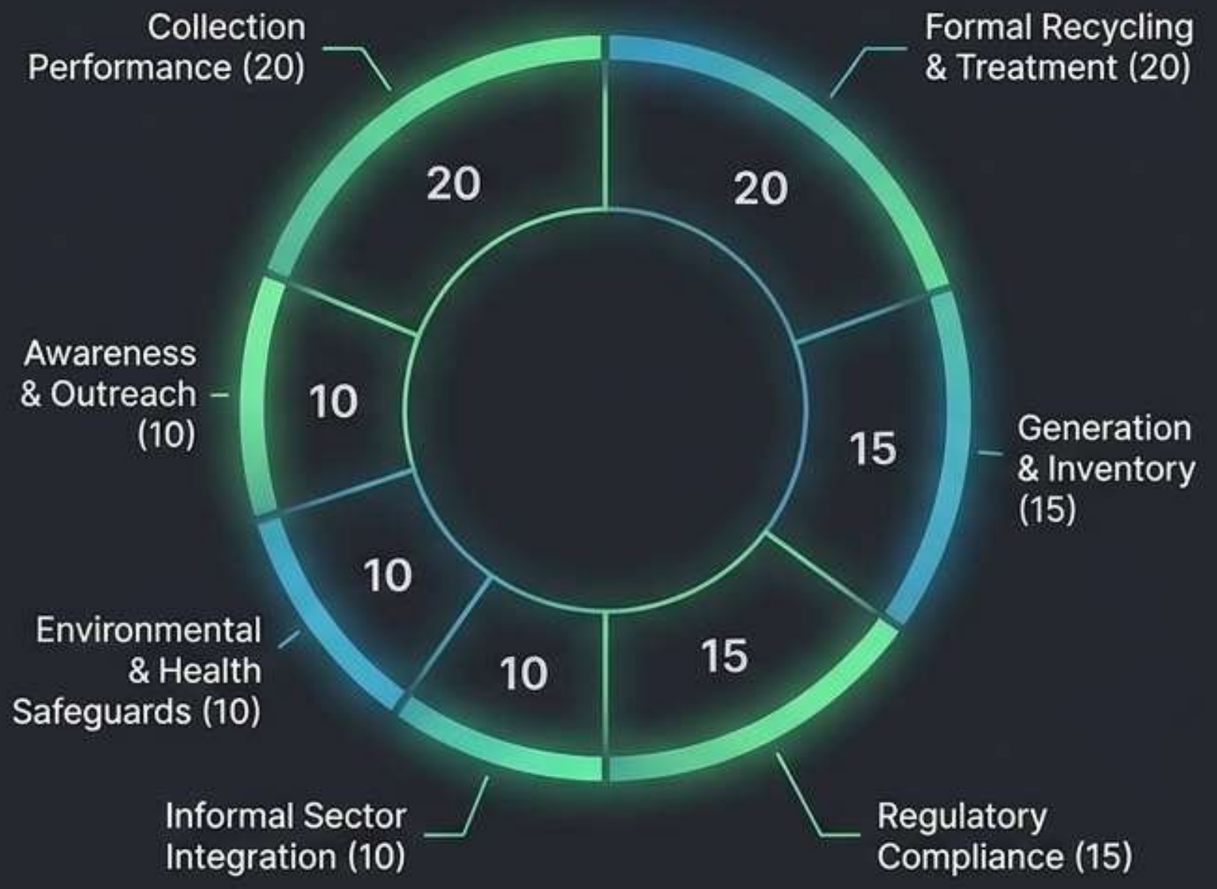
Zero Visibility into Priority Areas

The Solution Engine



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The algorithmic engine driving the state score



$$\text{Achievement \%} = \left(\frac{\text{Achieved Value}}{\text{Benchmark}}\right) \times 100$$

$$\text{Category Score} = \text{Achievement \%} \times \text{Weight}$$

$$\text{State Score} = \sum \text{Category Scores}$$

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Dashboard output reveals a critical system failure

42.3 / 100 RATING: CRITICAL

Score Bands Legend



85–100 (Excellent)	65–84 (Good)	45–64 (Needs Improvement)	0–44 (Critical)



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Diagnosing the gap: High treatment capacity, failed collection

Strengths

- Formal Recycling (87.7%)
- Health Safeguards (85.7%)
- Compliance (75%)

Weaknesses

- Generation (0%)
- Informal Sector (0%)
- Collection (7.1%)
- Awareness (35%)

Estimated Annual Generation:	Total Collected:	EPR Target:	EPR Achieved:
~3,032,600 kg	**~184,000 kg**	**214,600 kg**	**176,800 kg**

Intervention Priority: Collection performance and **informal-sector** formalisation.

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A centralized ecosystem for evidence-based planning

System Features

- Interactive KPI cards
- Automated score calculation
- Benchmark comparison
- Visual analytics & Performance tracking
- District-level analysis



Stakeholder Access

- State Pollution Control Boards (SPCB)
- Municipal corporations
- Government departments & Policymakers
- Environmental agencies & Researchers

Delivering a consolidated, evidence-based view for monitoring, planning, and corrective action.

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Roadmap for future enhancements and final conclusions

Future Enhancements Roadmap



- AI-based waste generation forecasting
- Machine learning using historical district data
- GIS-based hotspot mapping & IoT-enabled smart collection bins
- Citizen mobile reporting & National-level monitoring integration

The 42.3/100 (Critical) score reveals that while recycling infrastructure exists, inventory and collection require massive, immediate improvement.

“From fragmented data to actionable environmental intelligence.”

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